

Splat Regression Models

3DGS: solving an inverse problem, by fitting a splat model, using particle based optimization.

Splat Regression Models: a particle-based toolkit for low dimensional modeling.

Mean-field Dynamics

Setup: let $\mathcal{M}$ be a smooth Riemannian manifold, Haar msr. $\mathfrak{h}$
 $\mathcal{P}(\mathcal{M})$ measures on $\mathcal{M}$, $\mathcal{P}_{ac} = \{\mu \in \mathcal{P}(\mathcal{M}) : \mu \ll \mathfrak{h}\}$

Particles undergo motion in a drift field:

$$v : t, \mu_t \mapsto \text{vector field on } \mathcal{M}$$

abuse notation: $v(t, X, \mu) \in T_X(\mathcal{M})$